UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/570,060	01/06/2022	Yu Fang	SONA-0006-U01-C02	2220

87084 7590 06/27/2023

GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

EXAMINER	
ROY, SANJOY K	

ART UNIT	PAPER NUMBER
2443	

NOTIFICATION DATE	DELIVERY MODE
06/27/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 17/570,060	Applicant(s) Fang et al.	
	Examiner SANJOY K ROY	Art Unit 2443	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 01/06/2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-23 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-23 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 01/06/2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 01-06-2022.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. Claims 1-20 have been examined and rejected.

Claim Rejections - 35 USC § 102

3. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

4. Claims 1-20 are rejected under AIA 35 U.S.C. 102(a)(1) as being anticipated by Kim et al. (U.S. PGPub 2018/0072250).

As per claim1,

Kim teaches a system comprising: a mobile system having an Ethernet based network and a controller area network (CAN) based network, a CAN control device disposed onboard the mobile system, connected to the CAN based network, and structured to control operation of a component of the mobile system (KIM see para 0074, 0121, gateway 500 may be connected to a CAN-based vehicle network 20 include a plurality of CAN communication channels 530, plurality of CAN

communication channels 530 may be connected to the controller 510, fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30 In a step S830, when the verification of the security authentication request message is completed, the controller 510 may transmit a control command to the PHY layer 520, PHY layer 520 may activate the second port P2 connected to the target communication node 600 under control of the controller 510, step S962, the PHY layer 520 may transmit the diagnostic request message to the target communication node 600 through the second port P2 by mirroring, the PHY layer 520 may transmit the diagnostic request message to a plurality of communication nodes through a plurality of ports, target communication node 600 and the external diagnostic apparatus 10 may make a unicast address be included in a message generated after receiving the one-side forwarding request message, communication node 600 and the external diagnostic apparatus 10 may configure a destination of the message as a communication node, the message transmitted by the external diagnostic apparatus 10 may be delivered to the target communication node 600. The unicast address may include information that identifies the port to which the communication node, the destination of the message, is connected);

an Ethernet control device disposed onboard the mobile system, connected to the Ethernet based network, and structured to electrically communicate with the CAN control device(Kim see para 0054,0075, gateway 100 is connected with switches 110, 110-1, 110-2, 120, and 130, and supports a switch which supports an Ethernet protocol, and control at least one of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 connected to each of the switches where of the end nodes 111, 112, 113,

121, 122, 123, 131, 132, and 133 may include an electronic control unit ECU configured to control various types of devices mounted within a vehicle, gateway 500 include a plurality of communication ports P1, P2, P3, P4, and P5, first port P1 may be connected to an external diagnostic apparatus 1, second to fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30);

an Ethernet switch disposed onboard the mobile system and having a plurality of physical ports connected to the Ethernet based network, at least one physical port of the plurality forming part of a physical network path between the Ethernet switch and the Ethernet control device (Kim see para 0054,0075, gateway 100 is connected with switches 110, 110-1, 110-2, 120, and 130, and supports a switch which supports an Ethernet protocol, and control at least one of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 connected to each of the switches where of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 may include an electronic control unit ECU configured to control various types of devices mounted within a vehicle, gateway 500 include a plurality of communication ports P1, P2, P3, P4, and P5, first port P1 may be connected to an external diagnostic apparatus 1, second to fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30);

a CAN gateway disposed onboard the mobile system and connected to the CAN based network and the Ethernet switch(Kim see para 0074, gateway 500 may be connected to a CAN-based vehicle network 20 include a plurality of CAN

communication channels 530, plurality of CAN communication channels 530 may be connected to the controller 510, fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30);

and a network convergence circuit defined at least in part by the Ethernet switch and/or the CAN gateway, and structured to facilitate electronic communications between the Ethernet control device and the CAN control device by translating and relaying data messages between the Ethernet based network and the CAN based network (KIM see para 0074, 0121, gateway 500 may be connected to a CAN-based vehicle network 20 include a plurality of CAN communication channels 530, plurality of CAN communication channels 530 may be connected to the controller 510, fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30 In a step S830, when the verification of the security authentication request message is completed, the controller 510 may transmit a control command to the PHY layer 520, PHY layer 520 may activate the second port P2 connected to the target communication node 600 under control of the controller 510¸step S962, the PHY layer 520 may transmit the diagnostic request message to the target communication node 600 through the second port P2 by mirroring, the PHY layer 520 may transmit the diagnostic request message to a plurality of communication nodes through a plurality of ports, target communication node 600 and the external diagnostic apparatus 10 may make a unicast address be included in a message generated after receiving the one-side forwarding request message, communication node 600 and the external diagnostic apparatus 10 may configure a destination of the message as a communication node, the message

transmitted by the external diagnostic apparatus 10 may be delivered to the target communication node 600. The unicast address may include information that identifies the port to which the communication node, the destination of the message, is connected);.

As per claim 2

Kim teaches the system of claim 1, wherein the network convergence circuit is selectively configurable via a configuration command value transmitted over the Ethernet based network from a service device external to the mobile system (Kim see para 0102, 0104 fig. 9, in a step S910, the external diagnostic apparatus 10 may transmit a communication authentication request message for communications with the vehicle network, the PHY layer 520 may receive the communication authentication request message via the first port P1, the communication authentication request message may include identification information of the external diagnostic apparatus 10 and information on a diagnostic range of the external diagnostic apparatus 10, the PHY layer 520 may forward the communication authentication request message to the controller 510, S930, when the communication authentication for the external diagnostic apparatus 10 is completed, the external diagnostic apparatus 10 may transmit a vehicle network use request message. The PHY layer 520 may receive the vehicle network use request message through the first port P1. The PHY layer 520 may forward the vehicle network use request message to the controller 510).

As per claim 3

Kim teaches the system of claim 2, wherein: the plurality of physical ports includes a first physical port configured to selectively mirror another physical port of the plurality of physical ports in response to the configuration command value (Kim see para 0106, step S950, the controller 510 may configure the PHY layer so that the PHY layer performs port mirroring when it determines that the vehicle is in the stopped state. The controller 510 may cause a message received from the external diagnostic apparatus 10 to be mirrored to the target communication node 600, controller 510 may receive a diagnostic request message and configure the PHY layer 520 such that the message received from the first port P1 is mirrored to the second port P2, controller 510 may configure a destination of the message received from the first port P1 to the second port P2).

As per claim 4

Kim teaches the system of claim 2, wherein a configuration circuit is structured to selectively configure one or more portions of the network convergence circuit that are defined by the Ethernet switch and/or the CAN gateway(Kim see para 0111, 0120, step S962, the PHY layer 520 may transmit the diagnostic request message to the target communication node 600 through the second port P2 by mirroring, the PHY layer 520 may transmit the diagnostic request message to a plurality of communication nodes through a plurality of ports, target communication node 600 and the external diagnostic apparatus 10 may make a unicast address be included in a message

generated after receiving the one-side forwarding request message, communication node 600 and the external diagnostic apparatus 10 may configure a destination of the message as a communication node, the message transmitted by the external diagnostic apparatus 10 may be delivered to the target communication node 600. The unicast address may include information that identifies the port to which the communication node, the destination of the message, is connected).

As per claim 5

Kim teaches the system of claim 2 further comprising the service device, wherein the service device comprises: a testing circuit structured to communicate on the Ethernet based network, and generate a test command value that tests the CAN control device (Kim see para 0121, In a step S1050, the external diagnostic apparatus 10 may transmit a diagnostic request message and service time information. The diagnostic request message may include a unicast address indicating the target communication node 600, PHY layer 520 may receive the diagnostic request message and service time information, forward the diagnostic request message and the service time information to the controller 510, controller 510 may determine a port ID of the second port P2 to which the target communication node 600, which is the destination of the message, is connected from the unicast address included in the diagnostic request message, the unicast address may include an index key that identifies the destination of the message. The controller 510 may determine the communication node that matches the index key and determine an ID of a port to which the message is to be forwarded, controller 510 may determine the port ID of the port to which the diagnostic request

message is to be transmitted from the unicast address included in the diagnostic request message, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500)

As per claim 6

Kim teaches the system of claim 1, wherein the component of the mobile system relates to: an engine of the mobile system; a transmission of the mobile system; a braking system of the mobile system; a fuel system of the mobile system; and/or an electrical system of the mobile system(Kim see para 0055, 077, each of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 may include an electronic control unit ECU configured to control various types of devices mounted within a vehicle an infotainment device a display device, a navigation device, and an around view monitoring device, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500).

As per claims 7 and 12,

Kim teaches apparatus comprising: an Ethernet switch having a first network interface circuit that includes a plurality of physical ports each structured to interpret Ethernet based network data(Kim see para 0054,0075, gateway 100 is connected with switches 110, 110-1, 110-2, 120, and 130, and supports a switch which supports an Ethernet protocol, and control at least one of the end

nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 connected to each of the switches where of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 may include an electronic control unit ECU configured to control various types of devices mounted within a vehicle, gateway 500 include a plurality of communication ports P1, P2, P3, P4, and P5, first port P1 may be connected to an external diagnostic apparatus 1, second to fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30);

a translation circuit structured to determine a message value from the Ethernet based network data and to encode the message value in controller area network (CAN) data and a CAN gateway having a second network interface circuit structured to transmit the CAN data; wherein: the first network interface circuit, the translation circuit, and the second network interface circuit are defined by the Ethernet switch and/or the CAN gateway(KIM see para 0074, 0121, gateway 500 may be connected to a CAN-based vehicle network 20 include a plurality of CAN communication channels 530, plurality of CAN communication channels 530 may be connected to the controller 510, fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30 In a step S830, when the verification of the security authentication request message is completed, the controller 510 may transmit a control command to the PHY layer 520, PHY layer 520 may activate the second port P2 connected to the target communication node 600 under control of the controller 510, step S962, the PHY layer 520 may transmit the diagnostic request message to the target communication node 600 through the second port P2 by mirroring, the PHY layer 520 may transmit the diagnostic request message to a plurality

of communication nodes through a plurality of ports, target communication node 600 and the external diagnostic apparatus 10 may make a unicast address be included in a message generated after receiving the one-side forwarding request message, communication node 600 and the external diagnostic apparatus 10 may configure a destination of the message as a communication node, the message transmitted by the external diagnostic apparatus 10 may be delivered to the target communication node 600. The unicast address may include information that identifies the port to which the communication node, the destination of the message, is connected);

and the Ethernet Switch and the CAN gateway are both structured to be incorporated into a vehicle (Kim see para 0054,0075, gateway 100 is connected with switches 110, 110-1, 110-2, 120, and 130, and supports a switch which supports an Ethernet protocol, and control at least one of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 connected to each of the switches where of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 may include an electronic control unit ECU configured to control various types of devices mounted within a vehicle, gateway 500 include a plurality of communication ports P1, P2, P3, P4, and P5, first port P1 may be connected to an external diagnostic apparatus 1, second to fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30).

As per claims 8 and 13

Kim teaches the apparatus of claim 7 further comprising: a configuration circuit defined by the Ethernet switch and/or the CAN gateway and structured to modify the first network interface circuit, the translation circuit, and/or the second network interface circuit in response to a configuration command value(Kim see para 0102, 0104 fig. 9, in a step S910, the external diagnostic apparatus 10 may transmit a communication authentication request message for communications with the vehicle network, the PHY layer 520 may receive the communication authentication request message via the first port P1, the communication authentication request message may include identification information of the external diagnostic apparatus 10 and information on a diagnostic range of the external diagnostic apparatus 10, the PHY layer 520 may forward the communication authentication request message to the controller 510, S930, when the communication authentication for the external diagnostic apparatus 10 is completed, the external diagnostic apparatus 10 may transmit a vehicle network use request message. The PHY layer 520 may receive the vehicle network use request message through the first port P1. The PHY layer 520 may forward the vehicle network use request message to the controller 510).

As per claim 9

Kim teaches the apparatus of claim 8, wherein the configuration circuit is further structured to selectively configure a first physical port of the plurality selectively mirror a second physical port of the plurality in response to the configuration command value(Kim see para 0106, 0109 step S950, the controller 510 may configure the PHY layer so that the PHY layer performs port mirroring when it determines that the vehicle

is in the stopped state. The controller 510 may cause a message received from the external diagnostic apparatus 10 to be mirrored to the target communication node 600, controller 510 may receive a diagnostic request message and configure the PHY layer 520 such that the message received from the first port P1 is mirrored to the second port P2, controller 510 may configure a destination of the message received from the first port P1 to the second port P2, when there are a plurality of target communication nodes, a message received through the first port P1 may be mirrored so as to be transmitted through a plurality of ports p2 and p3).

As per claim 10

Kim teaches the apparatus of claim 9, wherein the configuration circuit is further structured to selectively configure a third physical port of the plurality selectively mirror the second physical port of the plurality in response to the configuration command value(Kim see para 0109 when there are a plurality of target communication nodes, a message received through the first port P1 may be mirrored so as to be transmitted through a plurality of ports p2 and p3) .

As per claim 11

1Kim teaches the apparatus of claim 8, wherein the configuration circuit is further structured to selectively configure which of one or more portions of the translation circuit are defined by the Ethernet switch and/or the CAN gateway(Kim see para 0111, 0120, step S962, the PHY layer 520 may transmit the diagnostic request message to the target communication node 600 through the second port P2 by mirroring, the PHY

layer 520 may transmit the diagnostic request message to a plurality of communication nodes through a plurality of ports, target communication node 600 and the external diagnostic apparatus 10 may make a unicast address be included in a message generated after receiving the one-side forwarding request message, communication node 600 and the external diagnostic apparatus 10 may configure a destination of the message as a communication node, the message transmitted by the external diagnostic apparatus 10 may be delivered to the target communication node 600. The unicast address may include information that identifies the port to which the communication node, the destination of the message, is connected).

As per claim 12

Kim teaches the method of claim 12 further comprising: interpreting a configuration command value; and modifying the translation circuit in response to the configuration command value (Kim see para 0106, step S950, the controller 510 may configure the PHY layer so that the PHY layer performs port mirroring when it determines that the vehicle is in the stopped state. The controller 510 may cause a message received from the external diagnostic apparatus 10 to be mirrored to the target communication node 600, controller 510 may receive a diagnostic request message and configure the PHY layer 520 such that the message received from the first port P1 is mirrored to the second port P2, controller 510 may configure a destination of the message received from the first port P1 to the second port P2).

As per claim 13

Kim teaches the method of claim 13, wherein modifying the translation circuit comprises: selectively configuring which of one or more portions of the translation circuit are defined at least in part by the Ethernet switch and/or the CAN gateway.

15. The method of claim 13 further comprising: generating the configuration command value external to the vehicle; and transmitting the configuration command value to the Ethernet switch (Kim see para 0102, 0104 fig. 9, in a step S910, the external diagnostic apparatus 10 may transmit a communication authentication request message for communications with the vehicle network, the PHY layer 520 may receive the communication authentication request message via the first port P1, the communication authentication request message may include identification information of the external diagnostic apparatus 10 and information on a diagnostic range of the external diagnostic apparatus 10, the PHY layer 520 may forward the communication authentication request message to the controller 510, S930, when the communication authentication for the external diagnostic apparatus 10 is completed, the external diagnostic apparatus 10 may transmit a vehicle network use request message. The PHY layer 520 may receive the vehicle network use request message through the first port P1. The PHY layer 520 may forward the vehicle network use request message to the controller 510).

As per claim 16

Kim teaches the method of claim 12 further comprising: generating a test command value external to the vehicle; transmitting the test command value to the Ethernet switch; and responsive to the test command value, executing a test procedure

involving a device on a CAN based network connected to the CAN gateway(Kim see para 0111, 0120, step S962, the PHY layer 520 may transmit the diagnostic request message to the target communication node 600 through the second port P2 by mirroring, the PHY layer 520 may transmit the diagnostic request message to a plurality of communication nodes through a plurality of ports, target communication node 600 and the external diagnostic apparatus 10 may make a unicast address be included in a message generated after receiving the one-side forwarding request message, communication node 600 and the external diagnostic apparatus 10 may configure a destination of the message as a communication node, the message transmitted by the external diagnostic apparatus 10 may be delivered to the target communication node 600. The unicast address may include information that identifies the port to which the communication node, the destination of the message, is connected).

As per claim 17

Kim teaches the method of claim 12, wherein transmitting, via a second interface circuit of the CAN gateway, the CAN based data set comprises: transmitting the CAN based data set to a vehicle control device on a CAN based network disposed onboard the vehicle, the vehicle control device controlling a component of the vehicle.

As per claim 18

Kim teaches the method of claim 17, wherein the component of the vehicle relates to: an engine of the vehicle; a transmission of the vehicle; a braking system of

the vehicle; a fuel system of the vehicle; and/or an electrical system of the vehicle (Kim see para 0055, 077, each of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 may include an electronic control unit ECU configured to control various types of devices mounted within a vehicle an infotainment device a display device, a navigation device, and an around view monitoring device, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500).

As per claim 19

Kim teaches the method of claim 12 further comprising: generating the Ethernet based data set via a vehicle control device on an Ethernet based network disposed onboard the vehicle, the vehicle control device controlling a component of the vehicle (Kim see para, 0113 step S980, the target communication node 600 may transmit the diagnostic response message. The gateway 500 may receive the diagnostic response message via the second port P2. The controller 510 may configure the PHY layer 520 such that the diagnostic response message received through the second port P2 is forwarded through the first port P1).

As per claim 20

Kim teaches the method of claim 19, wherein the component of the vehicle relates to: an engine of the vehicle; a transmission of the vehicle; a braking system of the vehicle; a fuel system of the vehicle; and/or an electrical system of the vehicle (Kim

see para 0055, 077, each of the end nodes 111, 112, 113, 121, 122, 123, 131, 132, and 133 may include an electronic control unit ECU configured to control various types of devices mounted within a vehicle an infotainment device a display device, a navigation device, and an around view monitoring device, external diagnostic apparatus 10 intends to diagnose an electrical component in the vehicle, the external diagnostic apparatus 10 may transmit a diagnostic request message to the gateway 500).

As per claim 21

Kim teaches the method of claim 12 further comprising down-sampling the Ethernet based data set (Kim see para 0122, controller 510 may determine a port ID of the second port P2 to which the target communication node 600, which is the destination of the message, is connected from the unicast address included in the diagnostic request message, the unicast address may include an index key that identifies the destination of the message. The controller 510 may determine the communication node that matches the index key and determine an ID of a port to which the message is to be forwarded, determine the port ID of the port to which the diagnostic request message is to be transmitted from the unicast address included in the diagnostic request message).

As per claim 22

Kim teaches the method of claim 21, wherein down-sampling the Ethernet based data set comprises configuring the Ethernet switch to perform a filtering function (Kim see para 0090, a step S822, the controller 510 may perform verification of the security

authentication request message to determine whether the diagnosis to be performed by the external diagnostic apparatus 10 is within an allowed security range. If the verification of the security authentication request message fails, the controller 510 may terminate the diagnostic procedure and may not allow communications between the external diagnostic apparatus 10 and the in-vehicle communication node).

As per claim 23

Kim teaches the method of claim 21, wherein down-sampling the Ethernet based data set comprises configuring the Ethernet switch to route traffic to a device to perform a filtering function (Kim see para 0091).

Conclusion

5. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. This includes:

U.S. PGPub 2018/0286152 Apparatus and method for converting protocol by type of data and vehicle system.

U.S. PGPub 2019/0020985 System and method for wireless interface selection and for communication and access control of subsystems, devices, and data in a vehicular environment

U.S. PGPub 2018/0287815 which describes an interface for connecting ethernet network with CAN network.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to examiner Sanjoy Roy, whose telephone number is 571-270-0675. The examiner can normally be reached on Mon-Fri, 8am.-5pm. (EST).

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Nicholas Taylor can be reached on 571-272-3889. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/SANJOY ROY/
Examiner, Art Unit 2443

/NICHOLAS R TAYLOR/
Supervisory Patent Examiner, Art Unit 2443